# Thesis checklist

Reassessed 8 November 2026 | Saved thesis, 22:33 | Outstanding items

This reassessment compares the existing checklist with the saved thesis. It contains only outstanding or deferred items; independent citation checking remains deferred until the final reference review. This is not a full proof audit.

## 3.5.3 - Energy functional

DEFERRED

**Review references at the end**

Check the Rudin bibliography mismatch and other references in the planned final review. The newly cited Ignatova entry is present in the saved bibliography.

Chapter 5: 624, 657; bibliography.bib: 1039-1047, 1133-1145.

OPEN

**Qualify numerical minimization**

One detected local minimum and the endpoints do not rule out competing minima. Explain how all candidates are found or qualify the selected value as a numerical estimate.

Chapter 5: 676-678; output/research/gaussian_qhat_properties_2026-10-29/report.md, section 2.

### Appendix - endpoint row-sum argument

OPEN

**Clarify endpoint bounds**

Explain why the local approximation and tail estimate apply when the limiting bandwidth is an endpoint. Refer to the full-lattice Riemann-sum estimate, rather than an interior-only finite-grid statement.

Appendix: 151-228, 230-258, 286-310.

### Chapter 5 - benchmark and conclusions

OPEN

**Document the benchmark and complete the table**

Give the bandwidth interval, geometric grid, timing scope, error reference and meaning of SE; add a descriptive caption. The project report describes a checked numerical reference, without an exhaustive certificate.

Chapter 5: 682-706; Code/Code_kde/outputs/brent_log_spectral_m100_comparison/report.md.

OPEN

**Restrict runtime and convergence conclusions**

Scope timing conclusions to the tested implementation and sample. State both lambda_N -> infinity and lambda_N/N^2 -> 0; avoid claiming monotone finite-N error or universal redundancy.

Chapter 5: 551, 709-713.

# Literature and polishing

Section 3.6 and final editorial review

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|---|---|
| OPEN | **Introduce the purpose and comparison criteria**<br>Explain how cited work relates to the thesis method, distinguish precedents from contributions, and add transitions between bandwidth selection, scale-space energy and Perron-vector literature.<br>Chapter 5: 761-805. |
| OPEN | **Clarify the Goldenshluger-Lepski presentation**<br>Use consistent notation for the risk criterion and selector, define the symbols and bandwidth domain, and make the wording and assumptions of the oracle inequality precise.<br>Chapter 5: 765-783. |
| OPEN | **Polish the scale-space and Perron-vector comparisons**<br>Introduce the empirical measure and change of notation in the scale-space calculation. Clarify the Shi and Baake comparisons; define rho and the relevant normalization in the localization statement.<br>Chapter 5: 784-799. |
| OPEN | **Justify the later localization claim**<br>The squared-Perron-weight claim uses a single h-star. State and use uniqueness, or formulate the claim for the minimizer set; in either case supply the concentration argument.<br>Chapter 5: 586-591, 801-805; appendix: 277-360. |
| OPEN | **Edit language and cross-references**<br>Correct grammar and terminology, improve paragraph flow, review formulas and figure references, and remove the Section 3.6 polishing marker after completing the review.<br>Chapter 5: 761-805. |
| OPEN | **General polishing**<br>Review Section 3.5.3 and the outlook after the mathematical changes: define symbols before use, correct typos, and review transitions and figure references.<br>Chapter 5: 599-759; outlook chapter. |

**Review basis:** Saved Chapter 5, appendix, outlook and bibliography; the cited project reports were used to check the local-minima formulas. Line references reflect the saved files at this reassessment.